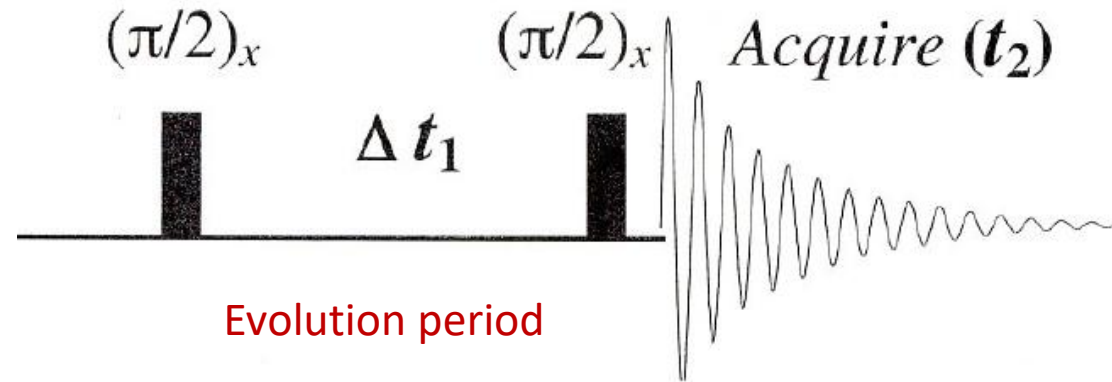


➤ **2D NMR:** Two time domains



- ❑ In the evolution period the spins are allowed to interact with each other

❑ *Correlation*: connectivity and coupling

- Same group of nuclei
- Nuclei which are coupled

❑ *Nuclear Overhauser effect (NOE)*: spatial proximity

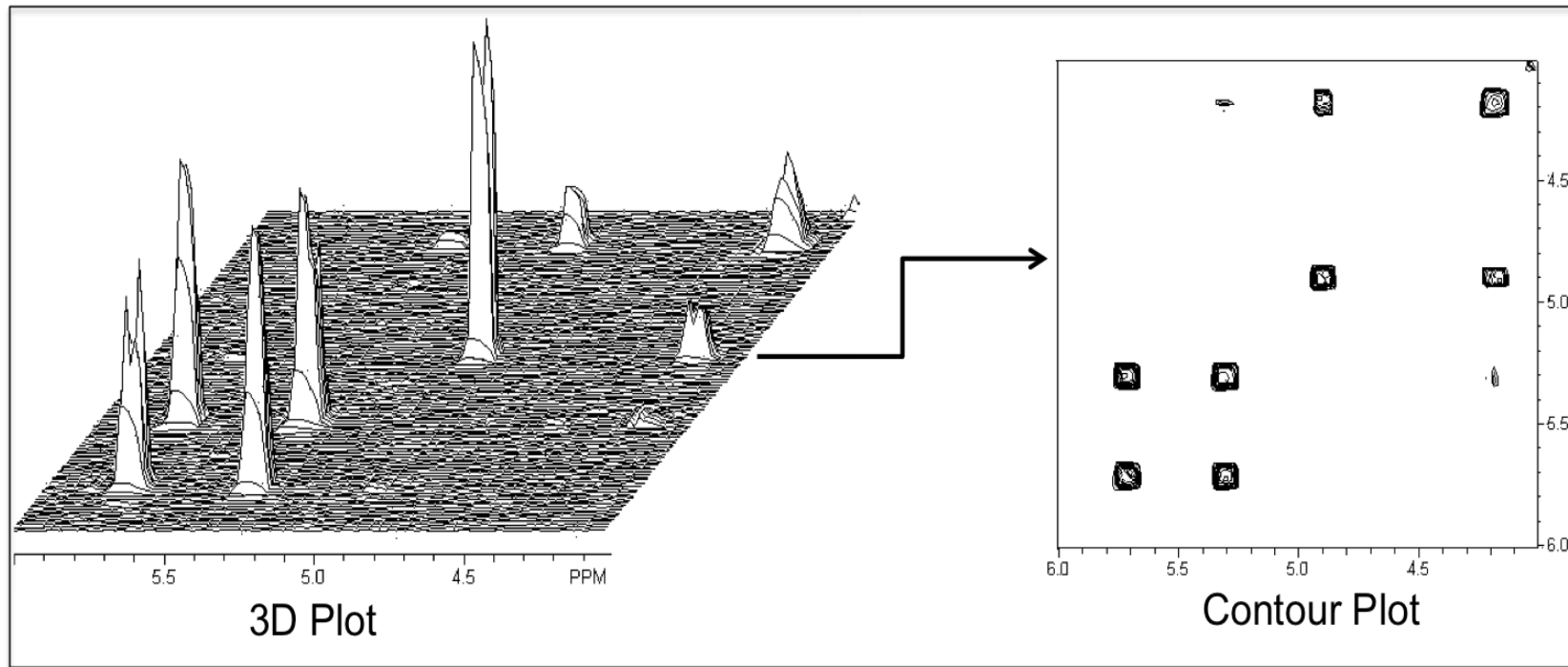
COSY: Correlation spectroscopy (J-coupling)

- ✓ ^1H - ^1H correlation via spin coupling interactions
- ✓ Nuclei must be connected by bonds (not too many bonds!)

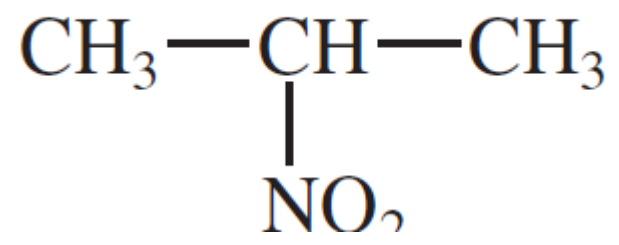
HETCOR: Heteronuclear correlation (J-coupling): ^1H - ^{13}C

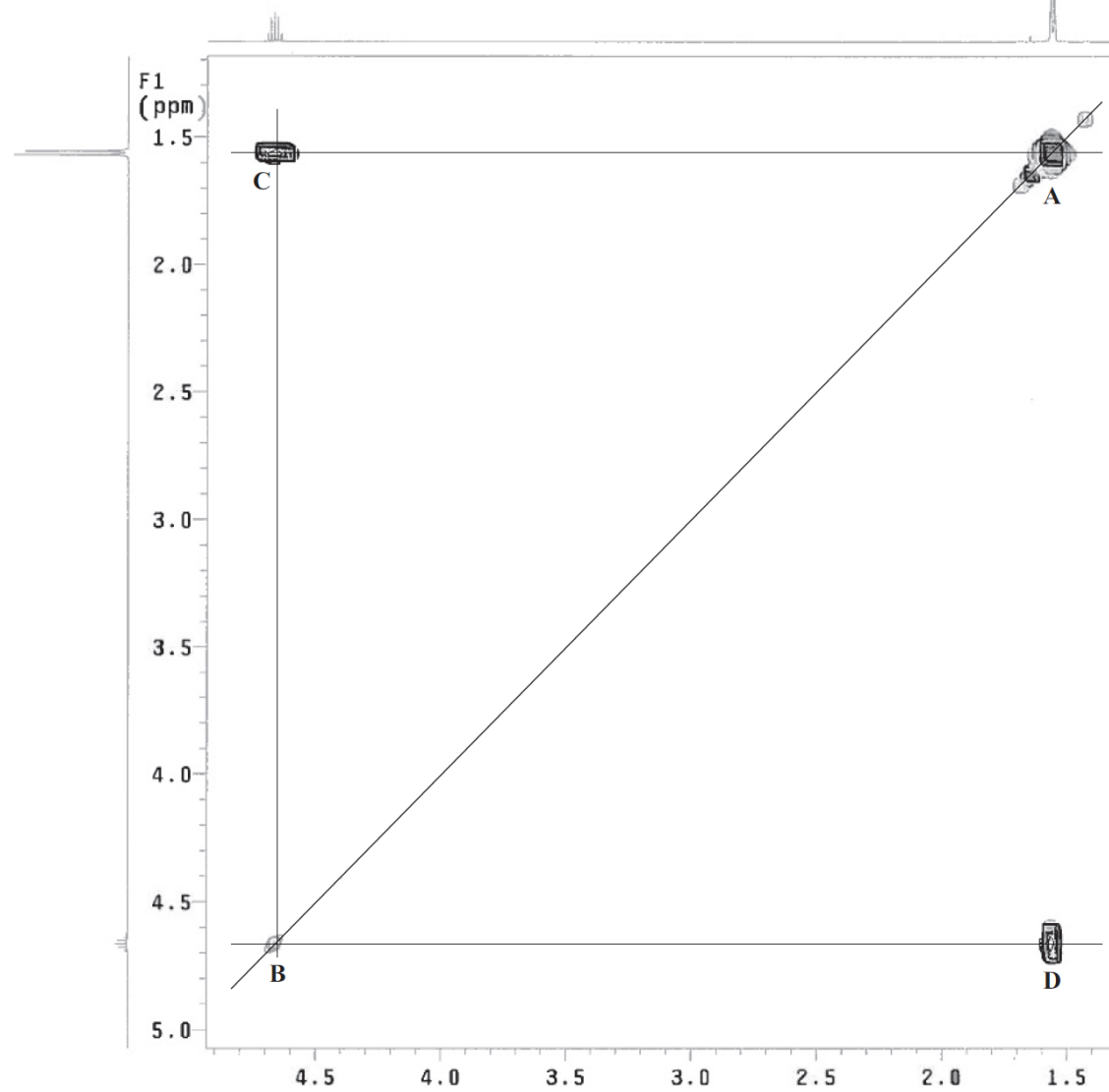
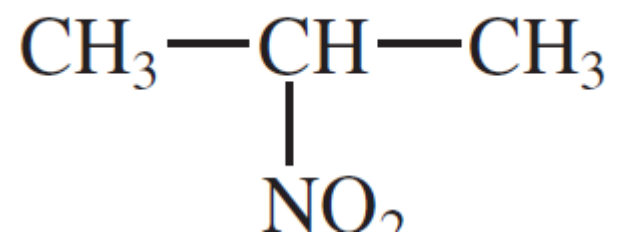
NOESY: NOE correlation, ^1H - ^1H correlation

- ✓ It correlates nuclei which are close to each other in space
- ✓ But not necessarily coupled to each other
- ✓ Basically works on the principle of NOE



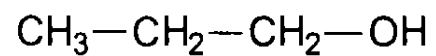
- *Diagonal peaks* have the same frequency coordinate on each axis and appear along the diagonal of the plot. Diagonal peaks correspond to the peaks in a 1D-NMR experiment
- *Cross peaks* have different values for each frequency coordinate and appear off the diagonal; they indicate couplings between pairs of nuclei (as splitting indicates couplings in 1D-NMR)



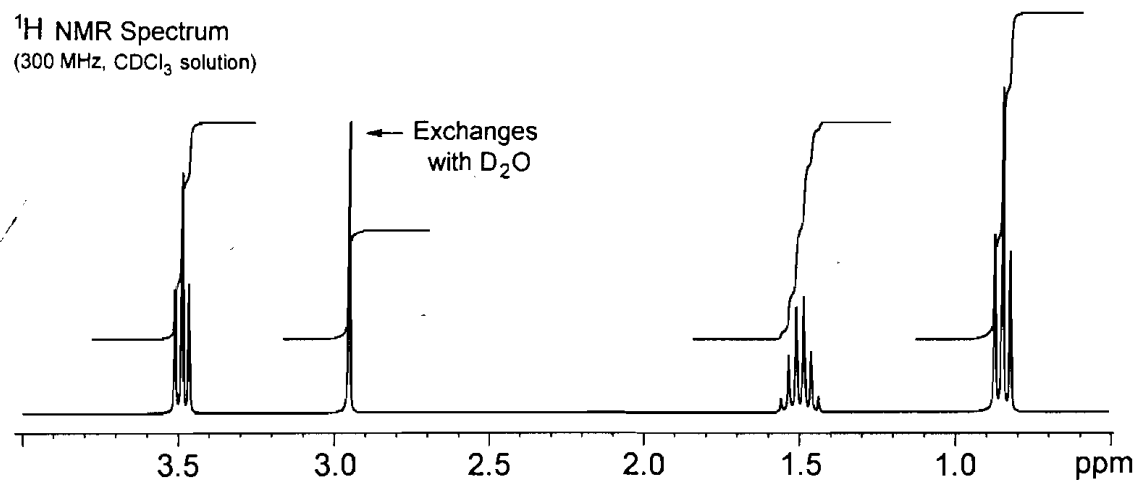


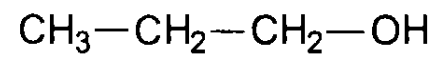
CORrelation SpectroscopY (COSY)

➤ COSY allows us to determine which peaks in a ^1H NMR spectra are coupled to which other peaks (what is splitting what)

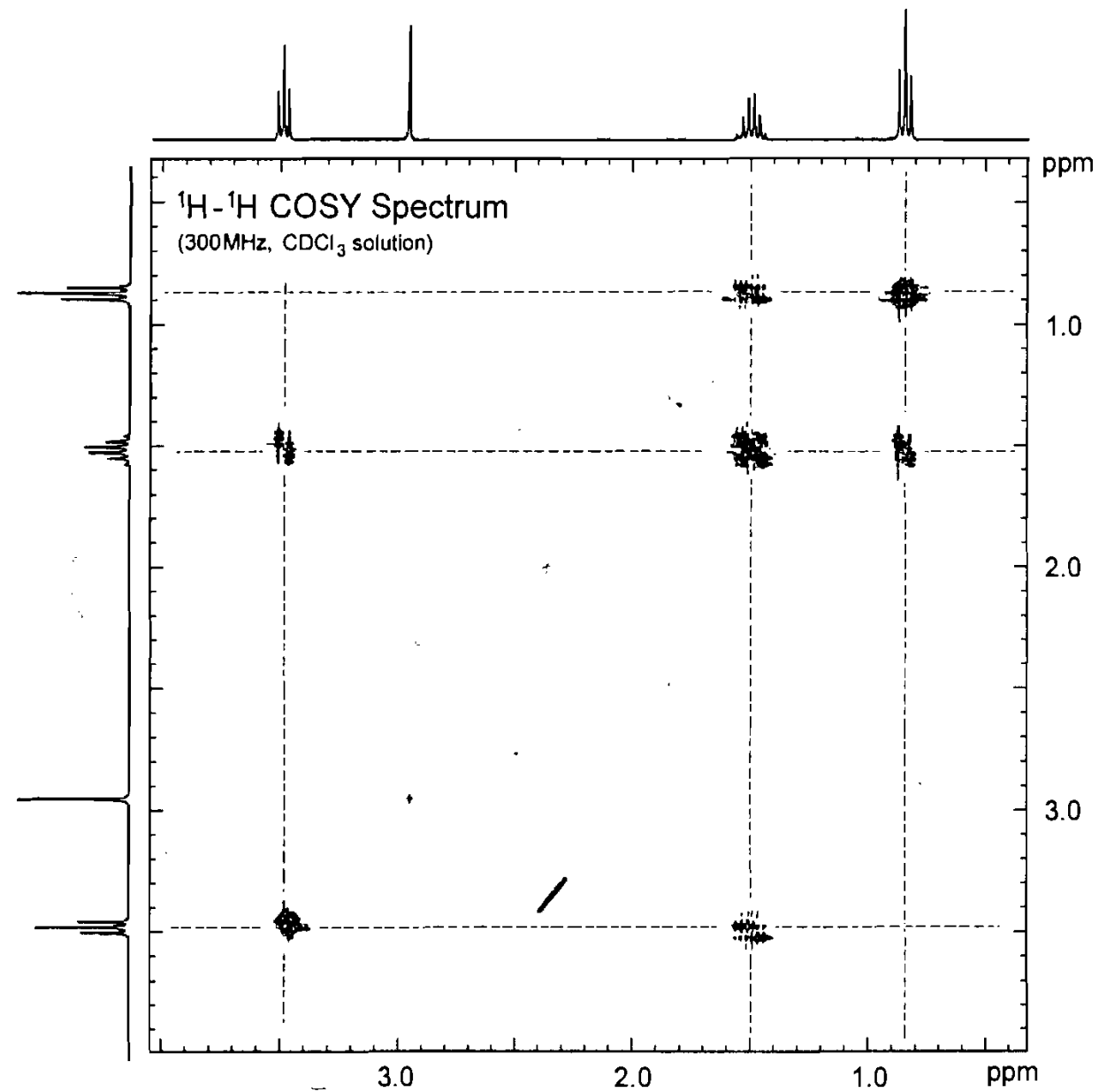
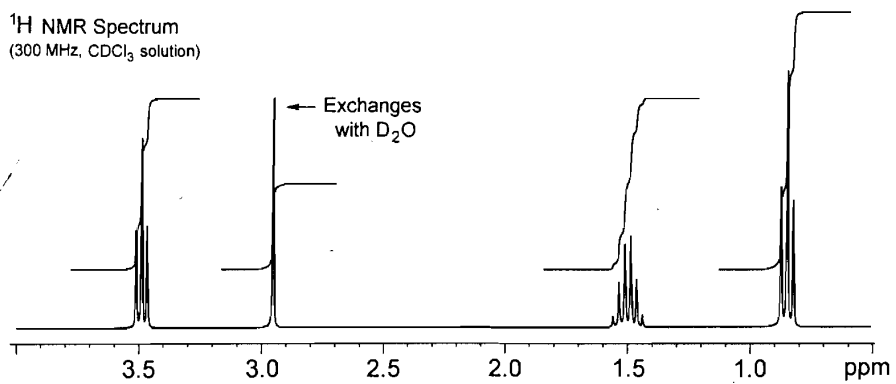


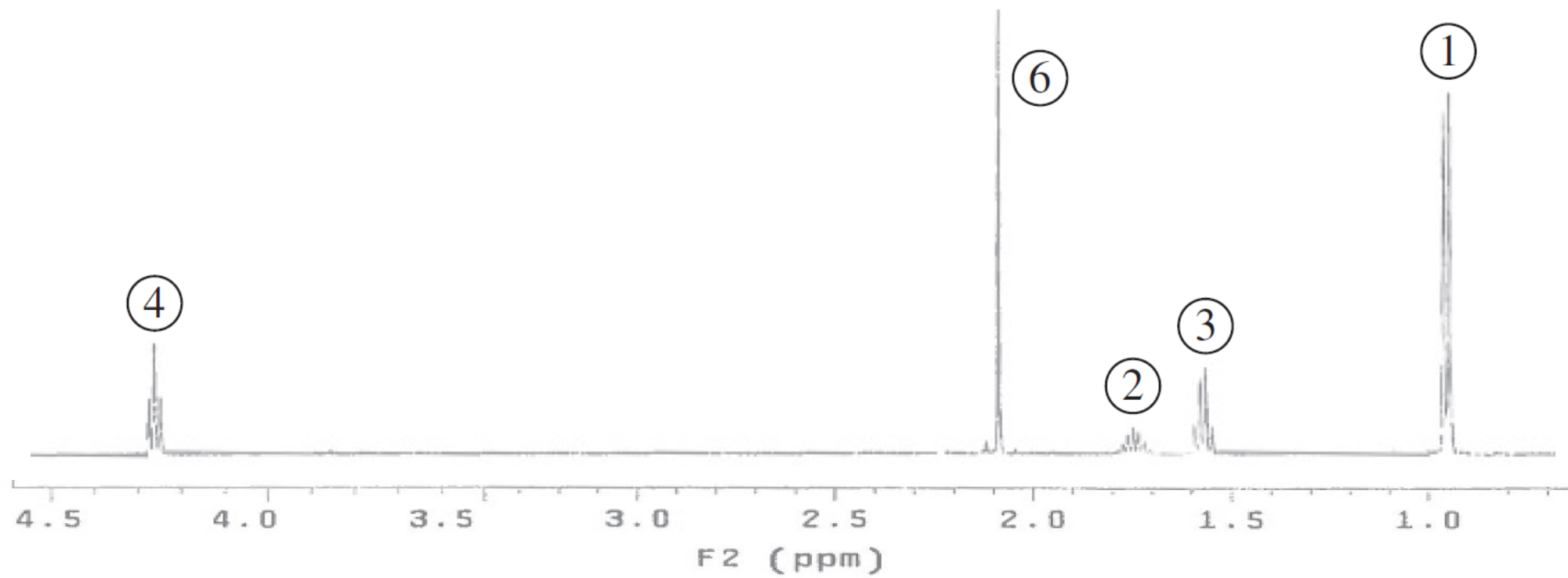
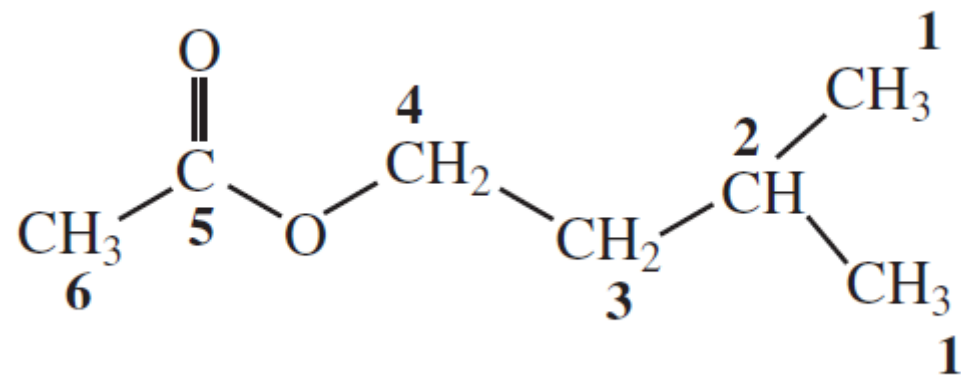
^1H NMR Spectrum
(300 MHz, CDCl_3 solution)

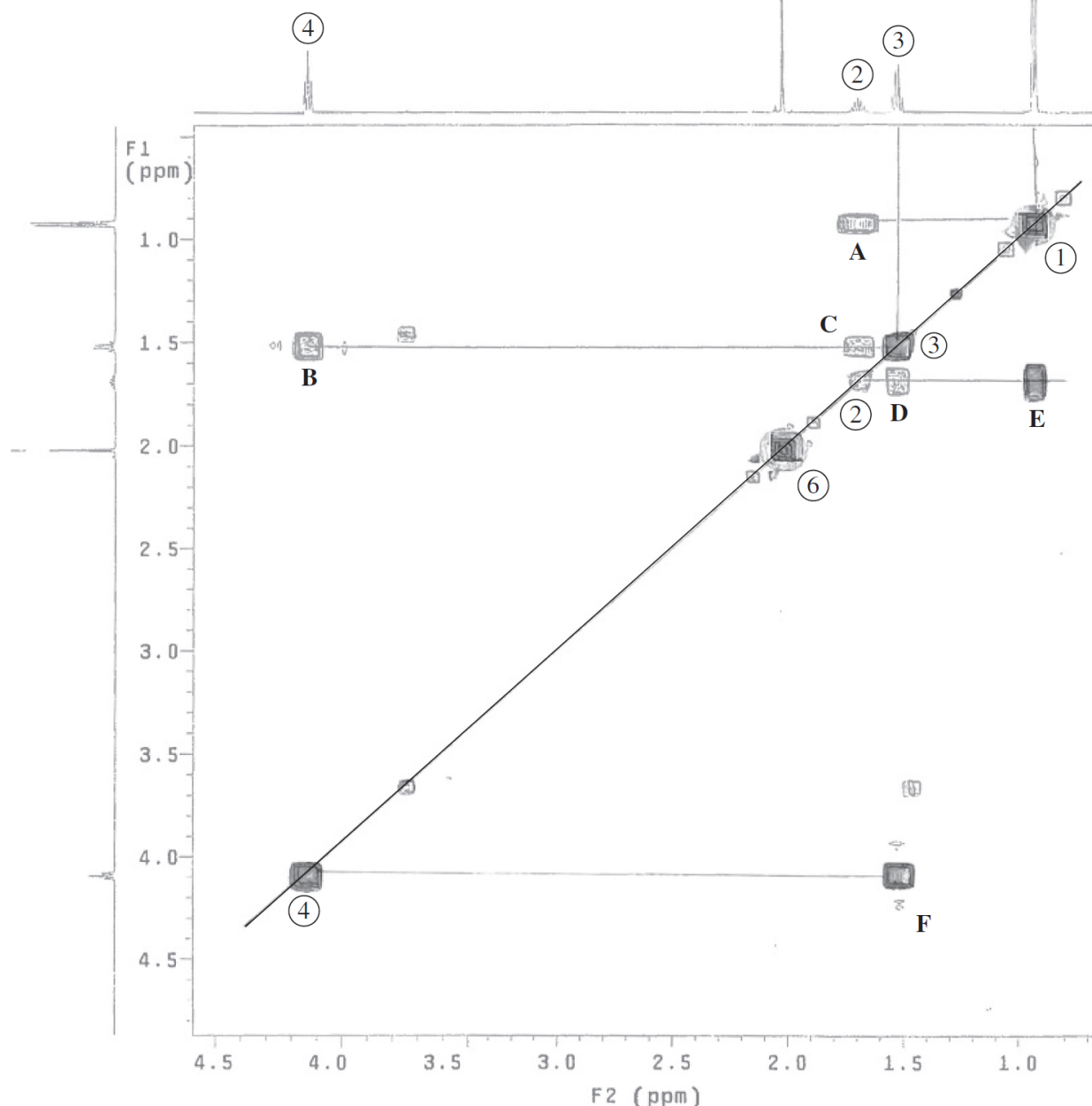
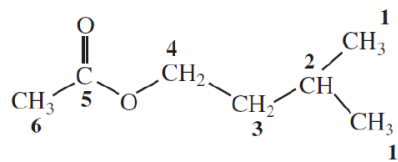




^1H NMR Spectrum
(300 MHz, CDCl_3 solution)

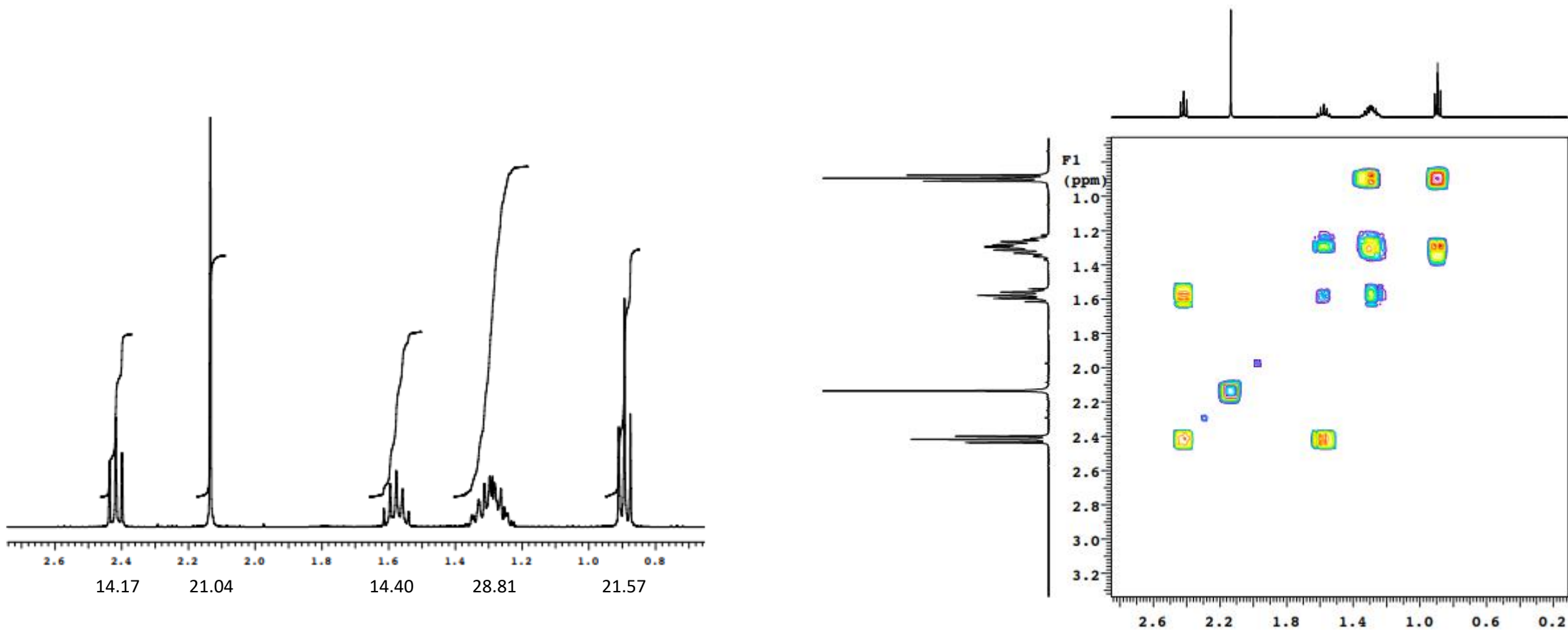






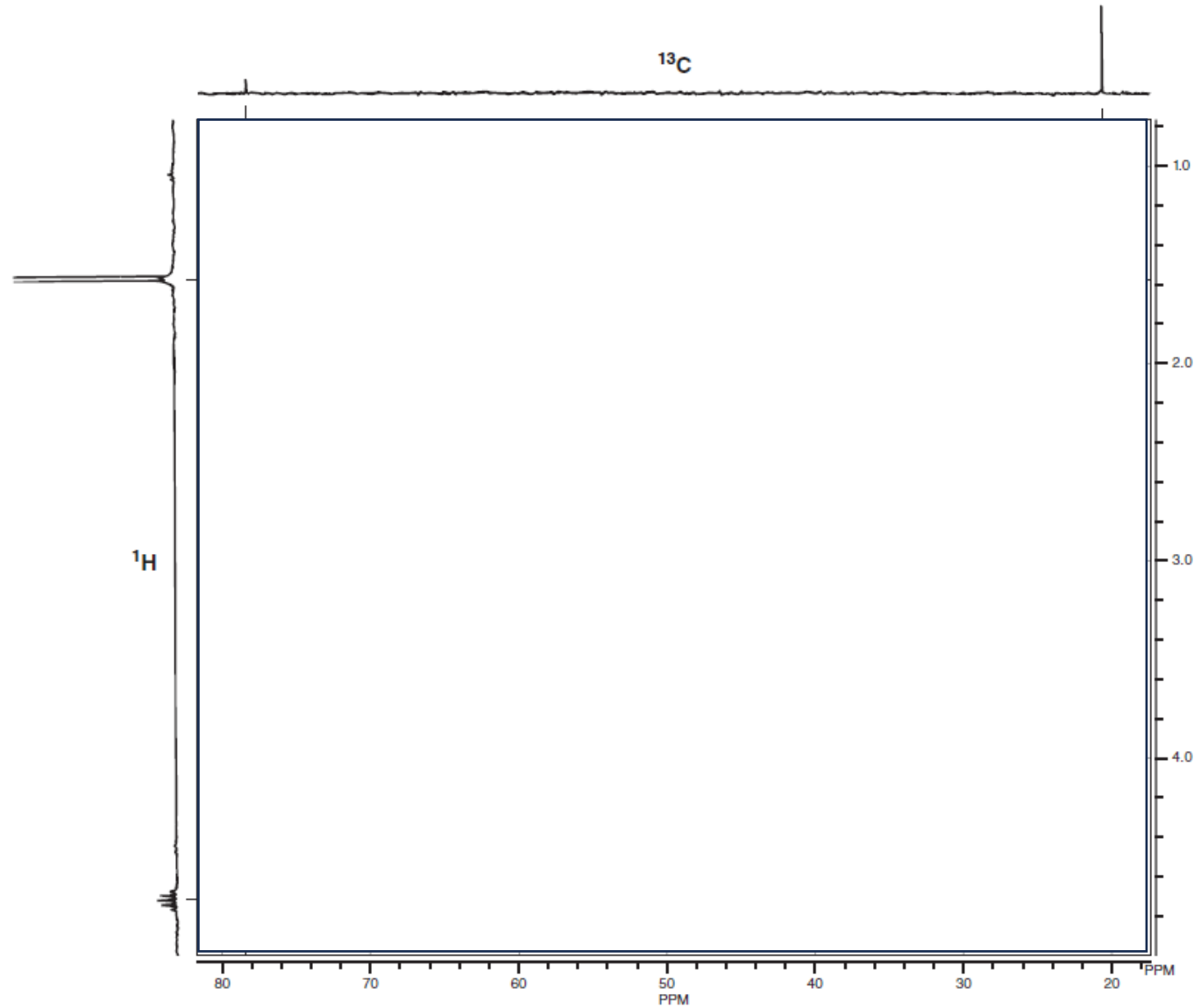
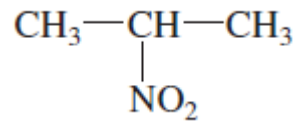
1. Predict the structure of the molecule based on the data given, the data doesn't indicate number of H, it indicates peak area of the protons. Explain the 2D NMR. The molecular formula of the compound is $C_7H_{14}O$.

^{13}C : 207, 43, 31, 29, 22.8, 22.4, 14.1



HETeronuclear **COR**relation (**HETCOR**)

^1H - ^{13}C correlations: ^1H NMR spectra on one axis and the ^{13}C NMR spectra on the other



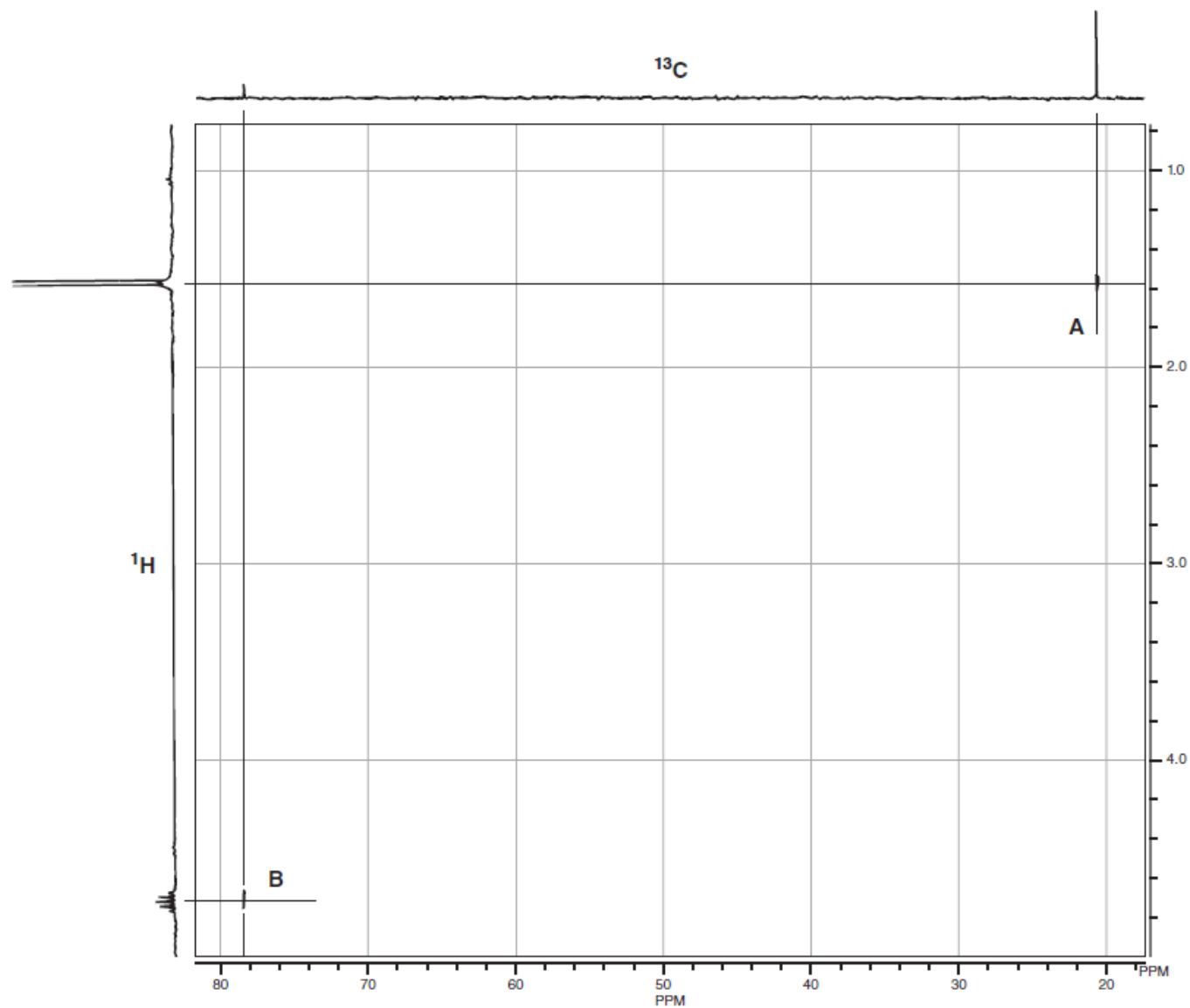
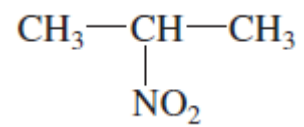
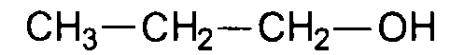
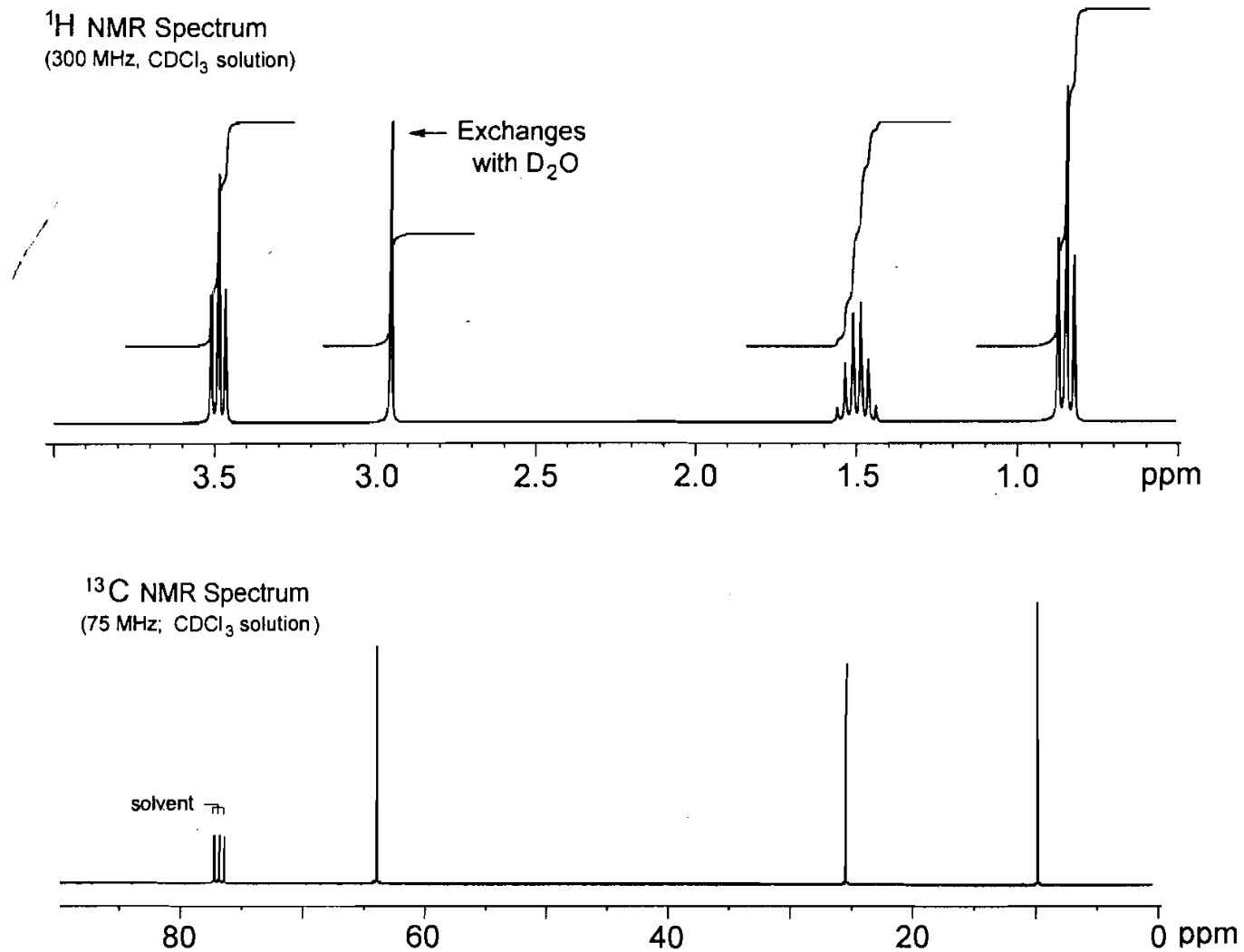
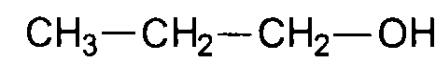
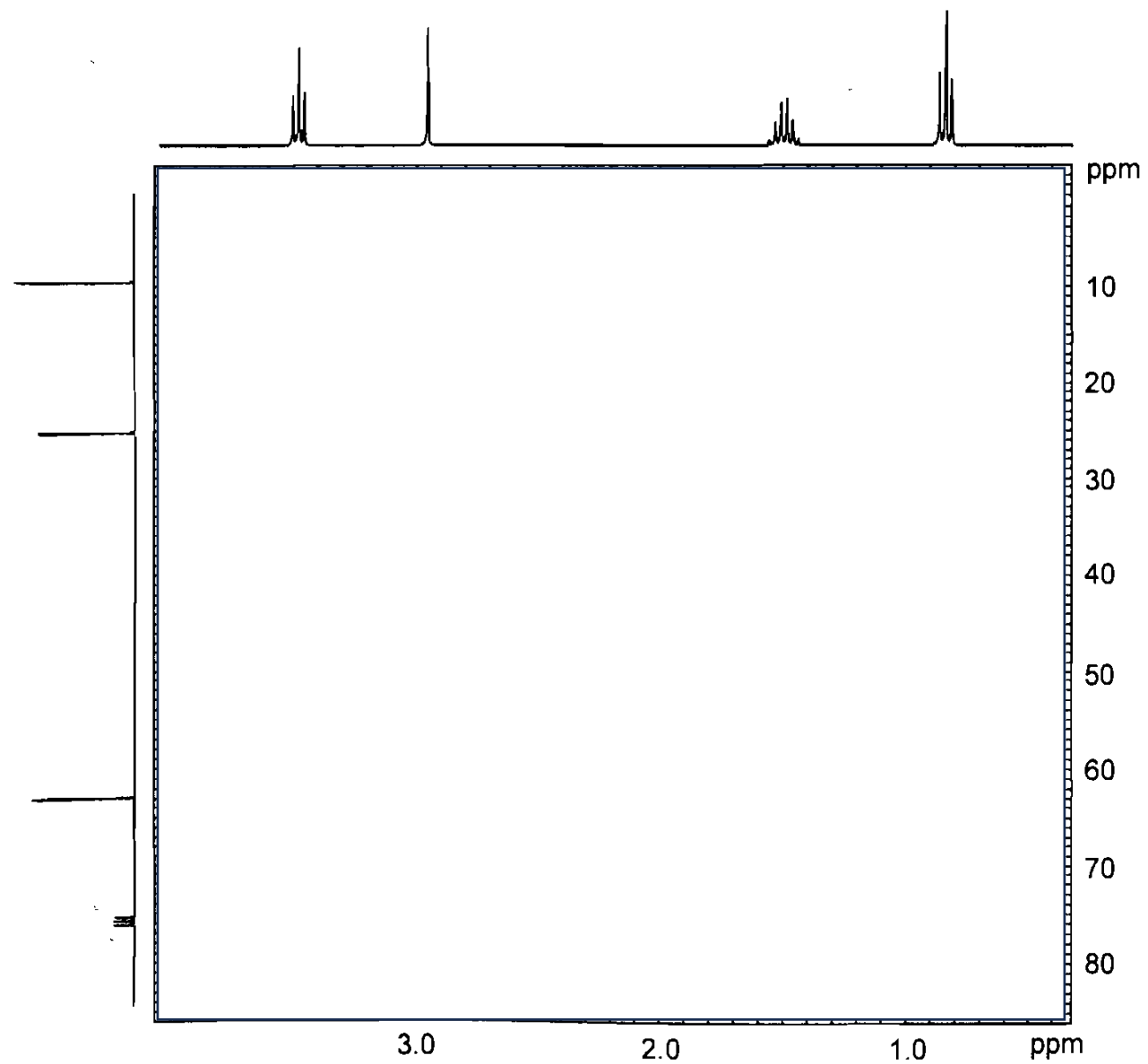


FIGURE 9.16 HETCOR spectrum of 2-nitropropane.

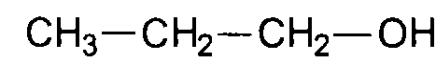
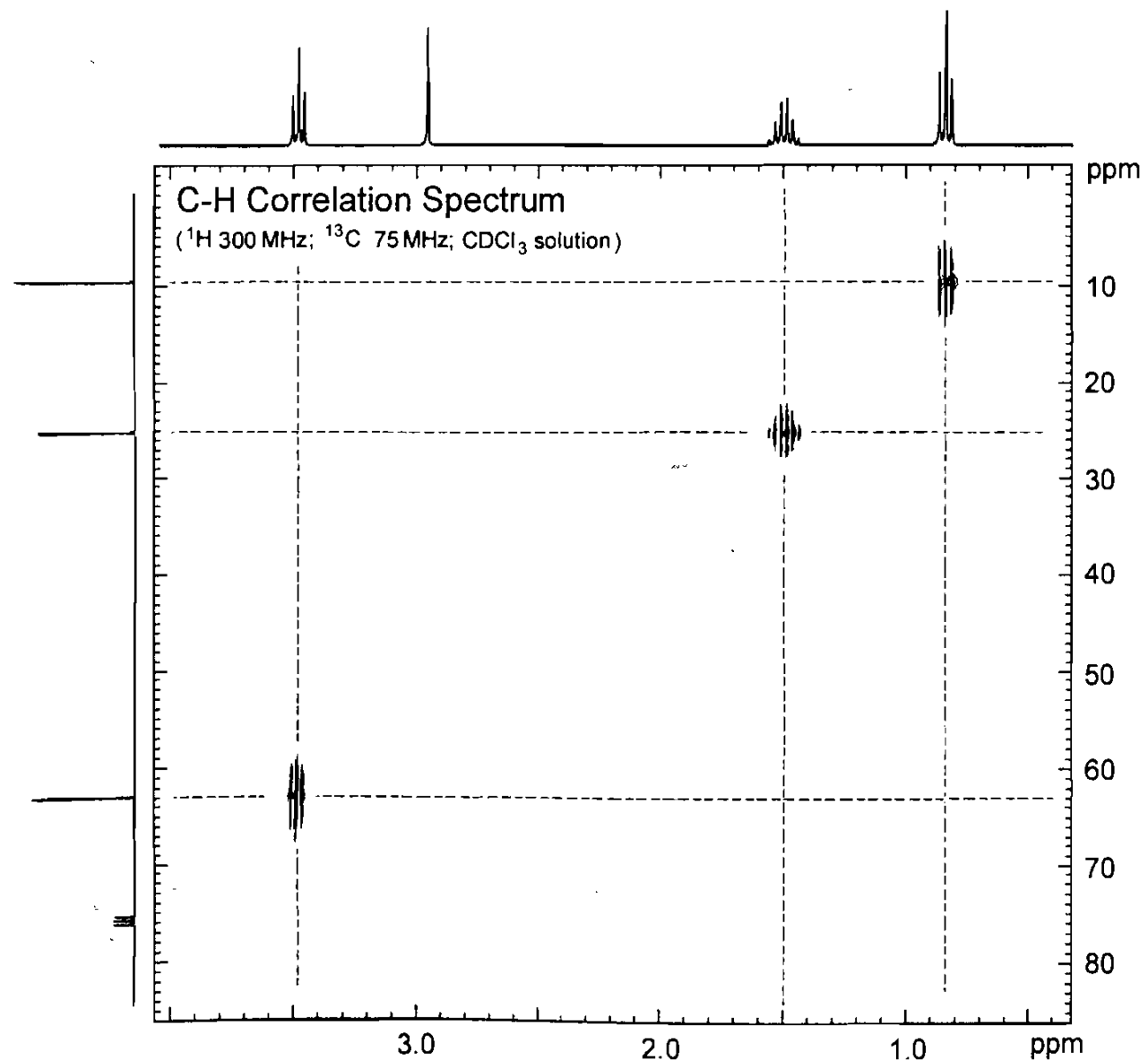
HETeronuclear **COR**relation (**HETCOR**)

^1H - ^{13}C correlations: ^1H NMR spectra on one axis and the ^{13}C NMR spectra on the other





No Diagonal peak



No Diagonal peak